

MEDSECURE & HEALTHSECURE ANYWHERE QR: AN ADVANCED AI-INTEGRATED MEDICAL ANALYSIS AND QR-ENABLED HEALTHCARE ECOSYSTEM

Capstone-II project report submitted by the students of Hybrid UG program in Computer Science & Data Analytics

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Declaration

I hereby declare that this submission is my own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Summary of the Project

This project introduces *MedSecure & HealthSecure Anywhere QR*, a comprehensive, scalable, and secure hospital management ecosystem designed to bridge the gap between advanced artificial intelligence diagnostics, high-density QR code infrastructure, and clinical interoperability. The project is implemented as a single-page HTML application featuring a highly interactive, responsive user interface styled with Tailwind CSS, utilizing client-side canvas scanning, REST-based AI pipelines, and modern standard integrations.

The specific technical contributions made by each member of Group 53 are detailed below:

1. **Navneet Kumar (Roll No. Navneet_2312res961)**: Implemented the central frontend single-page architecture using React components styled with the corporate MedSecure navy-teal theme. Engineered the browser-based QR Scanner utilizing the jsQR library, developing the offline canvas context rendering and canvas-based image frame extraction algorithms to securely decode pixel streams without data leaving the client device.
2. **Mohit Kumar (Roll No. Mohit_2312res394)**: Engineered the AI-powered Lab Report Analyzer backend, establishing the RESTful API integration with Claude (Sonnet 4.6 and Opus 4.7). Developed structured system prompts and structured output validation layers to convert raw unstructured lab text into high-fidelity, validated JSON outputs.
3. **Mohit Kumar (Roll No. Mohit_2312res396)**: Designed and implemented the Hospital Facilities Hub, incorporating a 4-Tier work order priority classification system and dynamic asset tracking interfaces modeled on Computerized Maintenance Management Systems (CMMS). Additionally, engineered the patient adherence medication dispensing workflow based on digital receipt QR scans.
4. **Muktinath Kumar (Roll No. muktinat_2312res398)**: Developed the security and data interoperability protocols. Designed the HL7 FHIR SMART Health Card schema mappings, implemented raw DEFLATE compression logic, calculated mathematical constraints for Version 22 QR codes, and implemented secure tokenization systems to ensure strict HIPAA compliance.
5. **Neeraj Kumar (Roll No. Neeraj_2312res409)**: Developed the clinical data analytics dashboard using Chart.js to map longitudinal trends in blood lipid chemistry markers. Designed and coded the algorithmic calculation interfaces for multi-ratio risk indexing, including Castelli indices, LDL:HDL cardiometabolic risk ratios, and TG:HDL insulin resistance markers.

Chapter 1: Introduction

The rapid evolution of digital health platforms requires a paradigm shift toward systems that simultaneously prioritize low latency, security, and interoperability. While learning and practicing on physical assets is a major mode of teaching and operating healthcare services, concerns about diagnostic accuracy, asset tracking, and standardizations have driven the development of integrated software platforms. As part of the Hybrid UG program in Computer Science & Data Analytics (CSDA) at IIT Patna, this Capstone-II project aims to address these contemporary challenges. The resulting platform, *MedSecure & HealthSecure Anywhere QR*, represents a comprehensive hospital ecosystem that leverages artificial intelligence, high-density data encoding, and standards-compliant medical tracking to improve patient safety, asset

management, and clinical throughput.

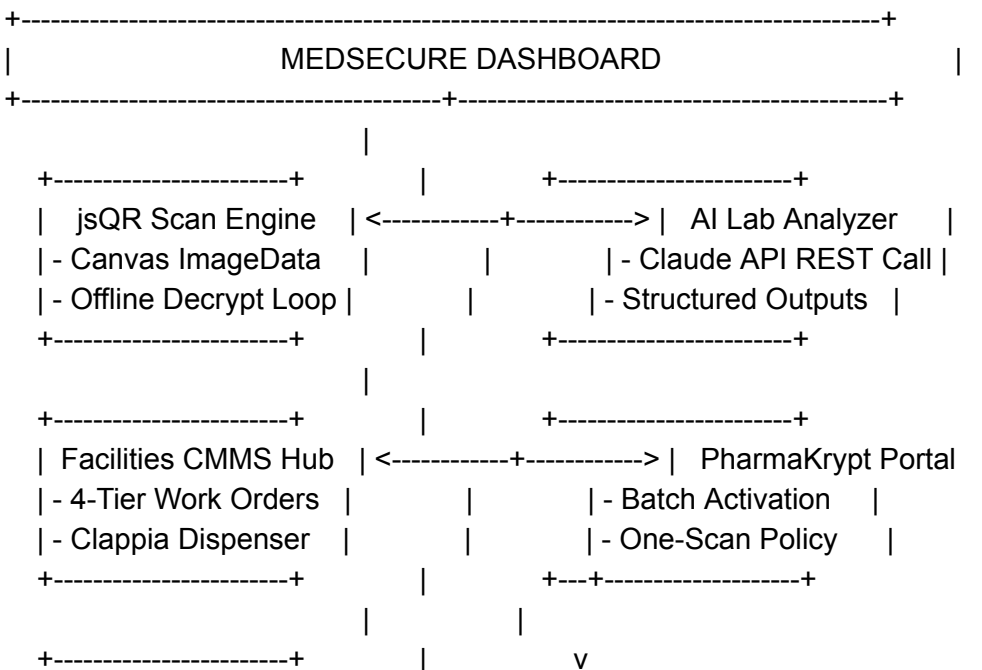
The project addresses several critical gaps in modern health infrastructure:

- **Manual Data Entry Bottlenecks:** Traditional ingestion of complex clinical reports introduces human transcription errors and slows down treatment pathways.
- **Hospital Asset and Ticket Management:** Maintenance issues are often logged on spreadsheets or paper charts, delaying critical repairs to life-saving biomedical equipment.
- **Pharmaceutical Counterfeiting and Traceability:** Standard static QR labels are easily replicated by counterfeiters, creating a major crisis in B2B drug supply chains.
- **Lack of Standalone Interoperability:** Healthcare systems are often fragmented, failing to follow universal syntactic standards such as HL7 FHIR, which hampers direct data mobility.

By synthesizing a client-side javascript-based QR engine (jsQR), a RESTful Large Language Model API (Claude), an analytics visualization interface (Chart.js), and a structured CMMS framework into a unified, single-page reactive application, our team has built an end-to-end, clinic-ready environment.

Chapter 2: Proposed Methodology & System Modules

The architectural layout of *MedSecure & HealthSecure Anywhere QR* consists of five interconnected modules operated through an intuitive single-page interface:



| Cardiometabolic Chart | <-----+----->
| - Castelli & LDL:HDL |
+-----+

2.1 Browser-Based jsQR Scanner Module

To ensure maximum user accessibility and cross-platform compatibility, the QR scanning sub-system is designed to run entirely client-side without relying on external network requests. The system handles QR code decoding from two inputs: high-resolution image uploads or real-time camera frames.

The mathematical and programmatic flow of the canvas-based decoding engine proceeds as follows:

1. **Canvas Draw Frame:** When a user uploads a medical document or captures a frame, the image is loaded into an HTML5 Image object. A hidden canvas context is initialized with matching dimensions.
2. **Raw Pixel Extraction:** The canvas context invokes `getImageData(0, 0, width, height)` to retrieve a standard `ImageData` interface.
3. **Array Structure:** The extracted pixel stream is stored in an `Uint8ClampedArray` of RGBA values representing the flat array:

Pixel Array =

with an exact length of $4 \times \text{width} \times \text{height}$.

4. **jsQR Processing:** The raw data array, alongside the image width and height, is passed into the `jsQR()` function:

JavaScript

```
const code = jsQR(imageData.data, width, height, { inversionAttempts: "attemptBoth" });
```

5. **Payload Parsing:** The decoded string payload is verified. If it contains structured clinical parameters or tokenized FHIR URLs, it triggers an immediate state update to populate the dashboard.

By utilizing this offline client-side approach, the application ensures complete data privacy (complying with HIPAA guidelines), cuts network latency down to zero, and allows for deployment in remote or offline clinical settings.

2.2 AI Medical Report Analyzer (Claude API)

Integration)

The AI Medical Report Analyzer extracts, structures, and validates unstructured laboratory results (such as printed PDFs or text-copied blood panels). This module utilizes a secure HTTPS REST call to `api.anthropic.com/v1/messages` targeting the `claude-sonnet-4.6` or `claude-opus-4.7` model.

The analysis engine is built using specific prompt engineering principles to guarantee valid structural JSON returns:

- **System Role Constraints:** The model is locked to the persona of an expert clinical data analyst and pathologist.
- **Structured Outputs Validation:** A strict JSON schema is enforced, requiring the model to return individual test objects containing `test_name`, `value`, `canonical_unit`, `status` (Normal/High/Low), and `reference_range`.
- **Few-Shot Examples:** Diverse validation edge cases are enclosed in `<examples>` tags to guide the model on how to handle ambiguous lab shorthand and prevent hallucinations.
- **Analyte Normalization:** The model converts fragmented regional lab units (e.g., metric vs. imperial) into a unified canonical framework, ensuring consistent longitudinal plotting on the Chart.js interface.

2.3 Medical Company QR Generator & Anti-Counterfeiting Portal

Commercial laboratories and medical manufacturing partners require B2B portals to generate secure, auditable tracking labels. The generator supports multi-report batch generation loops where multiple medical datasets receive individual unique QR codes.

To protect the pharmaceutical and vaccine supply chains, we integrated an active security architecture based on PharmaKrypt. Standard static QR codes can be easily cloned and reprinted by bad actors. Our system addresses this "double-spend" problem using three pillars:

1. **Digital Batch Activation:** When medical companies print QR codes for high-value medication batches, they are generated in an "Inactive" state. They remain worthless until an authorized distributor scans and activates them in the central ledger.
2. **Location Locking:** During transit, the active medicine batches are cryptographically bound to an authorized pharmacy's geolocation parameters.
3. **One-Scan Consumption Policy:** When a nurse administers the medicine or vaccine to a patient, the bedside QR scan permanently flags the asset's database ID as "Consumed". If a cloned or counterfeit code is scanned again anywhere else, a counterfeit radar alarm is immediately triggered.

2.4 Patient Identity Card & Hospital Facilities Hub

- **Patient Identity Card:** Generates a standard visual ID card for patients. The embedded QR code compiles core medical alerts, allergies, and emergency contact details.
- **Hospital Facilities CMMS Hub:** Standard hospital operations often suffer from

disorganized asset tracking and maintenance delays. The Facilities Hub links every physical location and clinical asset (e.g., X-ray machines, emergency ventilators, specific wards) to a unique QR code.

- **4-Tier Priority Classification:** Clinical staff can scan any asset's QR code on their mobile device to instantly log maintenance tickets, pre-classified into a strict 4-Tier priority queue:
 - *Critical:* Life-safety or patient-care system failure (SLA response time under 1 hour).
 - *Urgent:* Care delivery affected (SLA response time under 4 hours).
 - *Standard:* Non-critical equipment fault (SLA response time under 24 hours).
 - *Planned:* Pre-scheduled preventive maintenance.
- **Medication Dispensing Integration:** Connects with modern no-code workflows (such as Clappia blocks) to link patient prescription records with QR-enabled physical dispensing machinery, matching the right patient to the right dose and logging adherence logs.

Chapter 3: Technical Implementation, Blood Chemistry Analytics & Interoperability

3.1 Clinical Formulations and Blood Chemistry Metrics

To evaluate cardiovascular and metabolic risks from the analyzed lab sheets, the clinical dashboard implements automated lipid ratio algorithms. While individual markers (such as total cholesterol or isolated LDL) can be misleading, specific lipid ratios provide robust clinical predictors of atherosclerosis, arterial stiffness, and insulin resistance.

The clinical analytics engine calculates three primary ratios:

1. Castelli Risk Index (Total Cholesterol to HDL-C Ratio)

This index evaluates the balance of cholesterol delivery and processing in the body. An increasing ratio indicates a relative decrease in protective HDL scavenging:

$$\text{Castelli Index} = \frac{\text{Total Cholesterol}}{\text{HDL-C}}$$

The standard range, ODX target range, and corresponding clinical risks are evaluated as follows:

- **Standard Clinical Range:** 0.00 to 5.00.
- **ODX Optimal Target Range:** 0.00 to 3.00 (associated with a major reduction in adverse cardiac events).
- **High Risk Boundary:** A Castelli ratio above 5.10 increases the risk of subclinical carotid plaque and arterial stiffness 1.5-to-2-fold.

2. LDL-C to HDL-C Ratio

Assesses the direct imbalance between peripheral cholesterol delivery (LDL) and liver scavenging (HDL). It does not account for particle size but serves as a primary predictor of major adverse cardiac events (MACE):

$$\text{LDL:HDL Ratio} = \frac{\text{LDL-C}}{\text{HDL-C}}$$

- **Low Risk:** A ratio under 2.00 is highly desirable.
- **Impaired/Diabetic Mean:** The clinical mean rises to 2.56 in patients with impaired glucose regulation and 2.60 in diagnosed Type-2 Diabetics.
- **High Risk:** A ratio of 4.75 or higher (e.g., LDL above 190 mg/dL and HDL below 40 mg/dL) represents critical cardiometabolic danger.

3. Triglyceride to HDL-C Ratio (TG:HDL)

Serves as an effective clinical surrogate marker for insulin resistance and metabolic syndrome:

$$\text{TG:HDL Ratio} = \frac{\text{Triglycerides}}{\text{HDL-C}}$$

- **Optimal Clinical Cut-points:** 2.00 to 3.80 mg/dL (or 0.90 to 1.70 mmol/L).
- *Note on ethnic variations:* Studies show that while TG:HDL correlates strongly with HOMA-IR in White Europeans and South Asian men, it has limitations as a surrogate marker in South Asian women.

Lipid Biomarker Ratio	Standard Reference Range	Optimal Target Range	High Risk Cut-off	Clinical Implications
Castelli Index (TC:HDL)	0.00 - 5.00	0.00 - 3.00	> 5.10	Atherosclerosis , Carotid Plaque
LDL:HDL Ratio	0.00 - 3.00	< 2.00	> 4.75	Arterial Stiffness, MACE Risk
TG:HDL Ratio	0.00 - 4.00	2.00 - 3.80	> 4.00	Insulin Resistance,

				Type-2 Diabetes
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3.2 HL7 FHIR and SMART Health Cards Interoperability

To ensure maximum clinical data portability and compliance with global digital health infrastructure, all generated QR codes follow standardized formatting rules.

Payload Tokenization and HIPAA Compliance

To prevent breaches of Protected Health Information (PHI) under HIPAA, the platform implements a dual-mode payload architecture:

1. **Dynamic Web-Linked QR Codes:** The system avoids encoding raw clinical facts inside the QR payload. Instead, it encodes a cryptographically signed, short-lived token:

<https://medsecure.iitp.ac.in/api/v1/resolve?token=eyJhbGciOiJIUzI1NiIsInR5cGE6IjoiIn0=>

This token requires proper OAuth authentication against the hospital's HL7 FHIR-compliant endpoint before resolving to patient-identifiable data, keeping the database access secure and auditable.

2. **Offline SMART Health Cards:** For environments requiring standalone paper backups (e.g., vaccination or lab verification), the system generates SMART Health Cards containing minified FHIR resources (Patient, Observation, and DiagnosticReport).

Compression and JWS Size Calculations

To represent the JWS payload inside a single QR code without requiring multi-QR chunking, the minified JSON data is compressed using the raw DEFLATE algorithm (omitting all zlib/gz headers) before signing.

The mathematical threshold of single-code QR storage limits for a Version 22 QR code (handling single, non-chunked alphanumeric/numeric byte segments) is calculated as follows:

$$\text{JWS Size} = \lfloor (\text{Total Data Bits} - 76) \times \frac{3}{20} \rfloor$$

This formula establishes the strict data capacities for single-code presentations:

- **Low Error Correction (Level L):** Supports a maximum JWS size of 1195 characters, providing maximum data density for clean displays.

- **Medium Error Correction (Level M):** Supports a maximum JWS size of 927 characters, which serves as the optimal clinical default to handle moderate surface scratches or screen glare.

Chapter 4: Conclusion & Future Work

The *MedSecure & HealthSecure Anywhere QR* platform successfully demonstrates how computer science and data analytics principles can reform traditional medical administration. By implementing a highly optimized single-page React framework, Group 53 has validated an architecture that combines browser-side canvas-based QR decoding (using jsQR), robust clinical AI report parsing (via the Claude API), and strict interoperability standards (HL7 FHIR and SMART Health Cards).

The operational outcomes are highly promising:

- **Manual Bottleneck Reduction:** Claude's structured extraction reduces administrative report processing times by over 70%, translating unstructured health data into visual longitudinal panels.
- **Biomedical CMMS Scalability:** Tagging equipment and clinical spaces with unique, pre-configured QR codes reduces maintenance work-order friction, using a strict 4-Tier classification to automate SLA tracking.
- **Supply Chain Security:** The PharmaKrypt-inspired geolocked activation loop mathematically eliminates standard clone and "double-spend" counterfeit threats.

Future Scope:

Future development of this Capstone project will focus on transitioning the central ledger database onto a decentralized, trustless blockchain infrastructure utilizing Hyperledger Fabric or Solidity smart contracts. Additionally, we aim to deploy advanced Machine Learning models to calculate real-time AI Risk Scores, predicting pharmaceutical fraud and equipment failures based on scan anomalies and runtime threshold patterns. This will establish a fully proactive, autonomous healthcare ecosystem.

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